



Software
Performance
Engineering

LECTURE 3

Bentley Rules for optimizing Work

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Work

- **Definition.** The **work** of a program (on a given input) is the sum total of all the operations executed by the program.

The number of
executed instructions

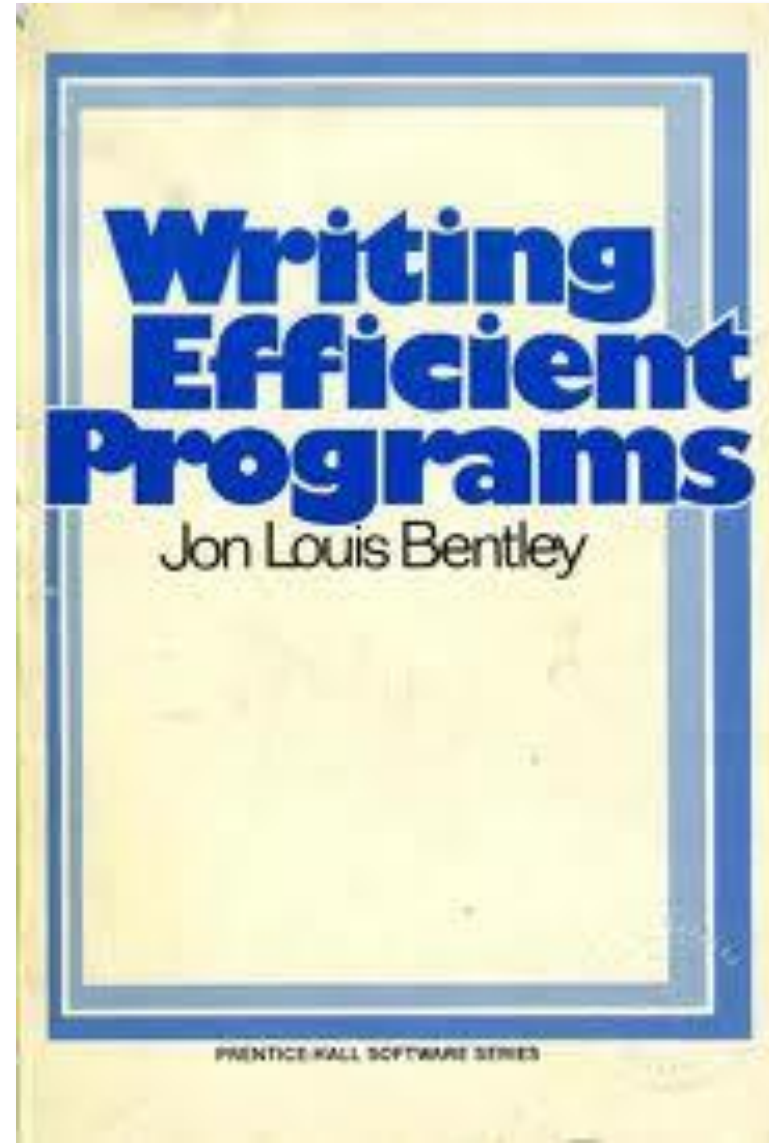


Avoid unnecessary
work

Reducing Work

- Less work $\approx$ faster code
- Reducing the work of a program does not automatically reduce its running time, due to the complex nature of computer hardware:
 - instruction-level parallelism (ILP),
 - caching,
 - vectorization,
 - speculation and branch prediction, etc
- But reducing work is a good heuristic for reducing overall exec. time
- Algorithm design can produce dramatic reductions in the work
 - e.g. when a $\Theta(n \lg n)$ -time sort replaces a $\Theta(n^2)$ -time sort

Jon Louis Bentley



1982

New Bentley Rules

Data structures

- Packing and encoding
- Augmentation
- Caching
- Precomputation
- Compile-time initialization
- Sparsity

Loops

- Loop unrolling
- Hoisting
- Loop fusion
- Eliminating wasted iterations

Logic

- Constant folding and propagation
- Common-subexpression elimination
- Algebraic identities
- Creating a fast path
- Short-circuiting
- Ordering tests
- Combining tests

Functions

- Inlining
- Tail-recursion elimination
- Coarsening recursion

Outline

- **Data structures**
- **Logic**
- **Loops**
- **Functions**

Packing and Encoding

Packing is to store more than one data value in a machine word. **Encoding** is to convert data into a representation that requires fewer bits

Example: Encoding dates

- The string “September 3, 2020” can be stored in 17 bytes — more than two 64-bit words — which must move whenever the date is manipulated.
- Assuming that we only store dates between 4096 B.C.E. and 4096 C.E., there are about $365.25 \times 8192 \approx 3\text{M}$ dates, which can be encoded in $\lceil \lg(3 \times 10^6) \rceil = 22$ bits, easily fitting in a 32-bit word.
- **Problem:** How can we represent dates compactly so that determining the year, month, and day is fast?

Packing and Encoding (2)

Example: Packing dates

- Let us pack the three fields into a word:

```
typedef struct {  
    int year: 13;  
    int month: 4;  
    int day: 5;  
} date_t;
```

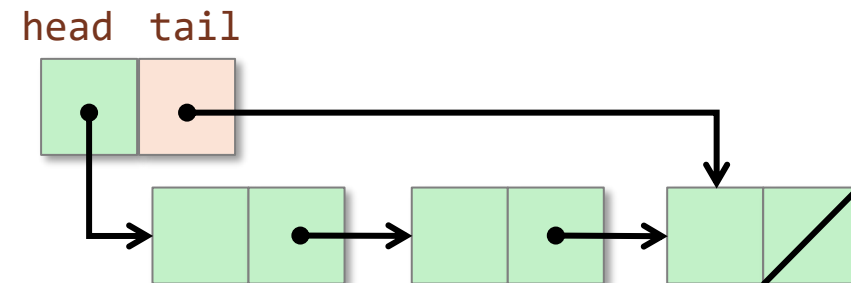
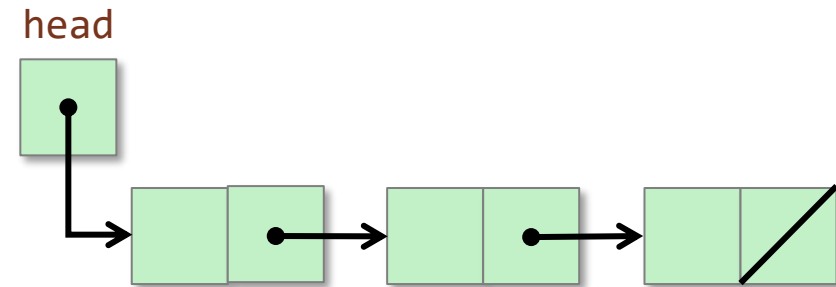
- Still only takes 22 bits
- But individual fields can be extracted much more quickly

Augmentation

- The idea is to **add information** to a data structure to make common operations do less work.

Example: Appending one singly linked list to another

- **Walk** through the first list, and set its null pointer to the start of the second
- **Augmenting** the list with a **tail** pointer allows appending to operate in constant time.



Caching

- The idea is to **store results** that have been accessed recently so that the program need not compute them again.

```
double hypotenuse(double A, double B) {  
    return sqrt(A*A + B*B);  
}
```

Before

About 30% faster
if cache is hit 2/3
of the time.

```
double cached_A = 0.0;  
double cached_B = 0.0;  
double cached_h = 0.0;  
  
double hypotenuse(double A, double B) {  
    if (A == cached_A && B == cached_B) {  
        return cached_h;  
    }  
    cached_A = A;  
    cached_B = B;  
    cached_h = sqrt(A*A + B*B);  
    return cached_h;  
}
```

After

Precomputation

- The idea is to **perform calculations in advance** so as to avoid doing them at “mission-critical” times.

Example: Binomial coefficients

$$\binom{n}{k} = \frac{n!}{k! (n - k)!}$$

- **Expensive** (lots of multiplications)
- Integer **overflow** for even modest values of n and k

Idea: Precompute the table of coefficients when initializing, and perform table look-up at runtime.

Step 1: Pascal's Triangle

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

1	0	0	0	0	0	0	0	0
1	1	0	0	0	0	0	0	0
1	2	1	0	0	0	0	0	0
1	3	3	1	0	0	0	0	0
1	4	6	4	1	0	0	0	0
1	5	10	10	5	1	0	0	0
1	6	15	20	15	6	1	0	0
1	7	21	35	35	21	7	1	0
1	8	28	56	70	56	28	8	1

```
int choose(int n, int k) {  
    if (n < k) return 0;  
    if (k == 0) return 1;  
    return choose(n-1, k-1) + choose(n-1, k);  
}
```

Step 2: Precomputing Pascal

```
#define CHOOSE_SIZE 100
int choose[CHOOSE_SIZE][CHOOSE_SIZE];

void init_choose() {
    for (int n = 0; n < CHOOSE_SIZE; ++n) {
        choose[n][0] = 1;
        choose[n][n] = 1;
    }
    for (int n = 1; n < CHOOSE_SIZE; ++n) {
        choose[0][n] = 0;
        for (int k = 1; k < n; ++k) {
            choose[n][k] = choose[n-1][k-1] + choose[n-1][k];
            choose[k][n] = 0;
        }
    }
}
```

Now, whenever we need a binomial coefficient (less than 100), we can simply index the **choose** array.

Compile-Time Initialization

- The idea is to **store** the values of **constants** during compilation, saving work at execution time.

Example

```
int choose[10][10] = {  
    { 1,  0,  0,  0,  0,  0,  0,  0,  0,  0, },  
    { 1,  1,  0,  0,  0,  0,  0,  0,  0,  0, },  
    { 1,  2,  1,  0,  0,  0,  0,  0,  0,  0, },  
    { 1,  3,  3,  1,  0,  0,  0,  0,  0,  0, },  
    { 1,  4,  6,  4,  1,  0,  0,  0,  0,  0, },  
    { 1,  5, 10, 10,  5,  1,  0,  0,  0,  0, },  
    { 1,  6, 15, 20, 15,  6,  1,  0,  0,  0, },  
    { 1,  7, 21, 35, 35, 21,  7,  1,  0,  0, },  
    { 1,  8, 28, 56, 70, 56, 28,  8,  1,  0, },  
    { 1,  9, 36, 84, 126, 126, 84, 36,  9,  1, },  
};
```

Compile-Time Initialization (2)

- **Idea:** Create large static tables by **metaprogramming**.

```
#define N 100
int main(int argc, const char *argv[]) {
    init_choose();
    printf("#define N %3d\n", N);
    printf("int choose[N][N] = {\n");
    for (int a = 0; a < N; ++a) {
        printf("    {");
        for (int b = 0; b < N; ++b) {
            printf("%3d, ", choose[a][b]);
        }
        printf("},\n");
    }
    printf("};\n");
}
```

Sparsity

- The idea is to **avoid** storing and computing on **zeroes**.
- “The fastest way to compute is not to compute at all.”

Example: Matrix-vector multiplication

$$y = \begin{pmatrix} 3 & 0 & 0 & 0 & 1 & 0 \\ 0 & 4 & 1 & 0 & 5 & 9 \\ 0 & 0 & 0 & 2 & 0 & 6 \\ 5 & 0 & 0 & 3 & 0 & 0 \\ 5 & 0 & 0 & 0 & 8 & 0 \\ 0 & 0 & 0 & 9 & 7 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 4 \\ 2 \\ 8 \\ 5 \\ 7 \end{pmatrix}$$

Dense matrix-vector multiplication performs $n^2 = 36$ scalar multiplies, but only **14** entries are nonzero.

Sparsity

- The idea is to **avoid** storing and computing on **zeroes**.
- “The fastest way to compute is not to compute at all.”

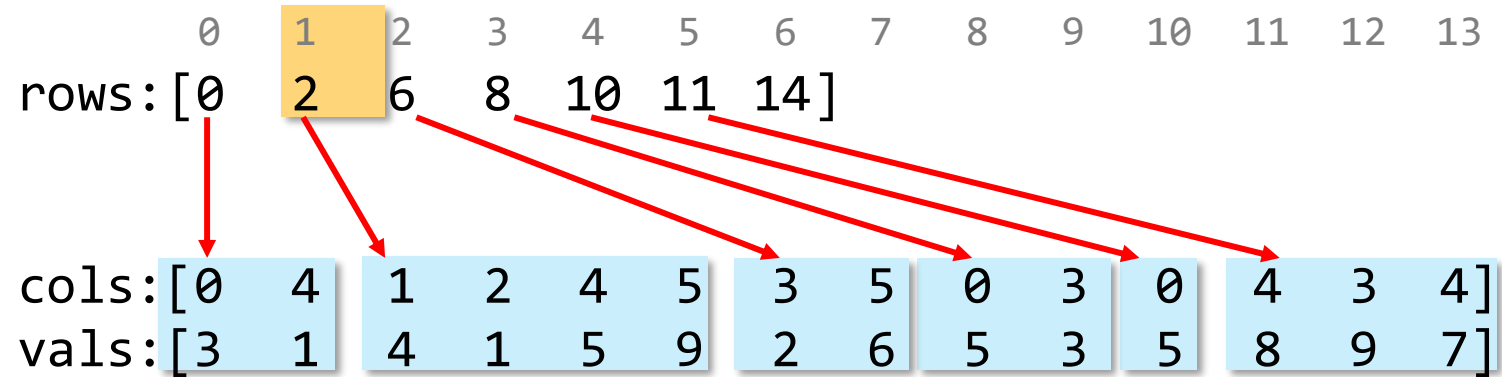
Example: Matrix-vector multiplication

$$y = \begin{pmatrix} 3 & & & & 1 & & \\ & 4 & 1 & & 5 & 9 & \\ & & & 2 & & 6 & \\ 5 & & & 3 & & & \\ 5 & & & & 8 & & \\ & & & 9 & 7 & & \end{pmatrix} \begin{pmatrix} 1 \\ 4 \\ 2 \\ 8 \\ 5 \\ 7 \end{pmatrix}$$

Dense matrix-vector multiplication performs $n^2 = 36$ scalar multiplies, but only **14** entries are nonzero.

Sparsity (2)

Compressed Sparse Rows (CSR)



0	☐	3	0	0	0	1	0	[
1	☐	0	4	1	0	5	9	[
2	☐	0	0	0	2	0	6	[
3	☐	5	0	0	3	0	0	[
4	☐	0	0	0	0	5	0	[
5	☐	0	0	0	8	9	7	[
		0	1	2	3	4	5	

$n = 6$
 $nnz = 14$

Storage is $O(n+nnz)$ instead of n^2

Sparsity (3)

CSR matrix-vector multiplication

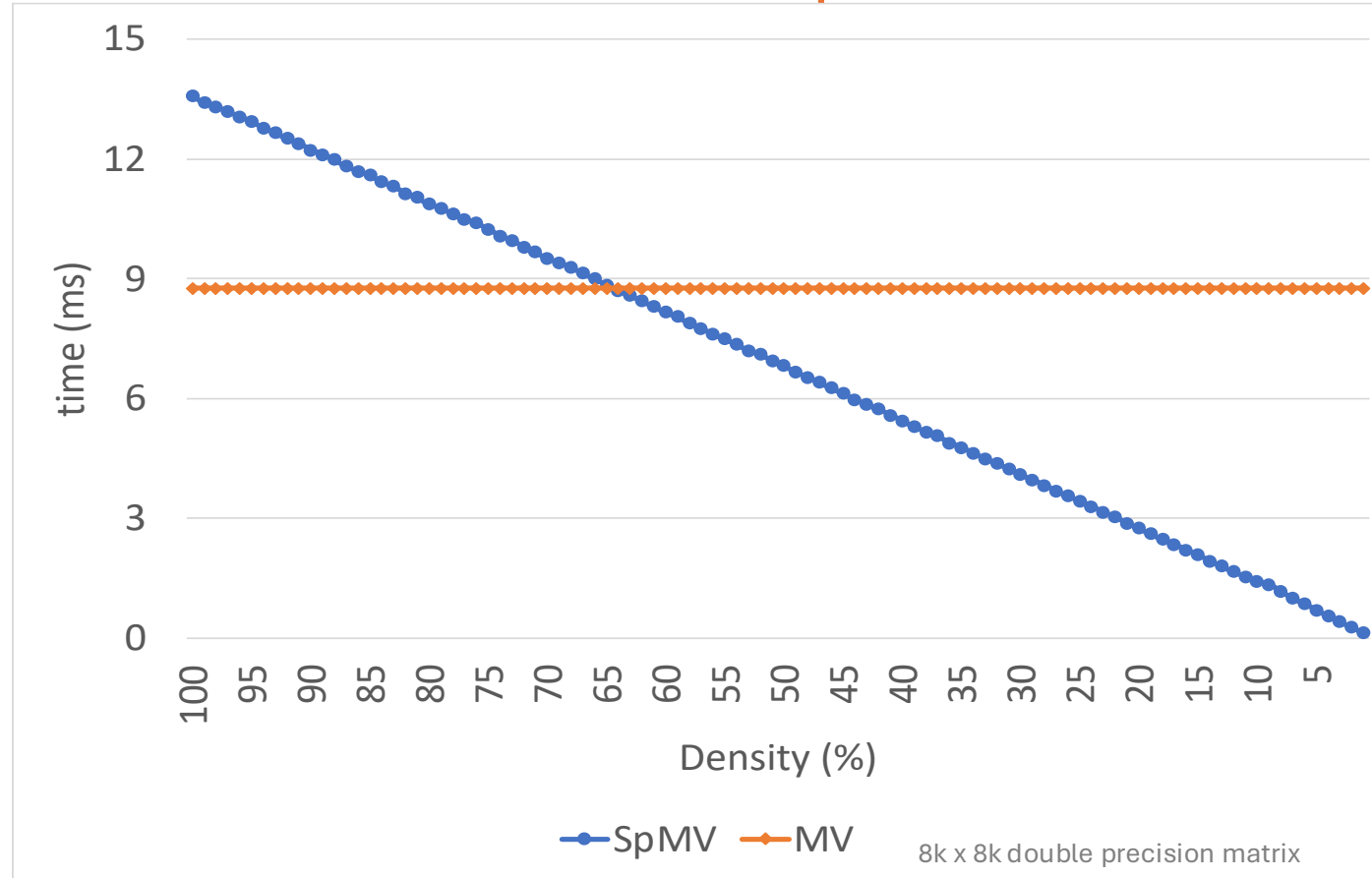
```
typedef struct {
    int n, nnz;
    int *rows;    // length n
    int *cols;    // length nnz
    double *vals; // length nnz
} sparse_matrix_t;

void spmv(sparse_matrix_t *A, double *x, double *y) {
    for (int i = 0; i < A->n; i++) {
        y[i] = 0;
        for (int k = A->rows[i]; k < A->rows[i+1]; k++) {
            int j = A->cols[k];
            y[i] += A->vals[k] * x[j];
        }
    }
}
```

Number of scalar multiplications = **nnz**, which is potentially much less than **n^2** .

Sparsity (3)

CSR matrix-vector multiplication

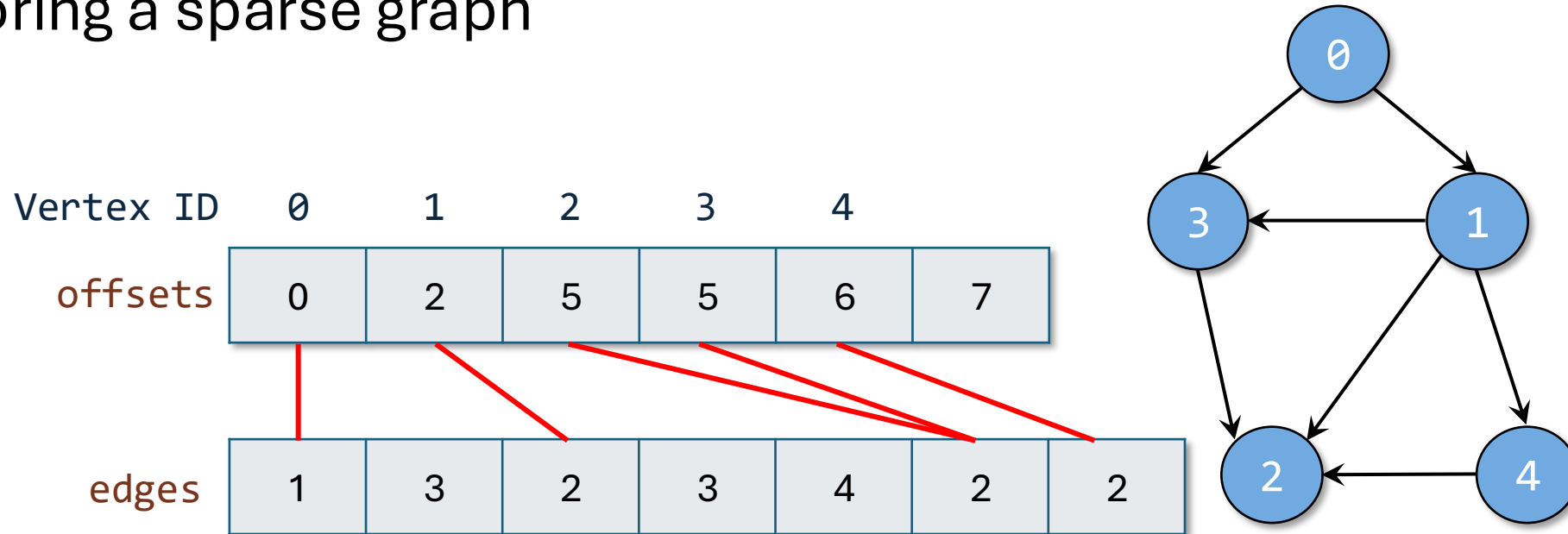


Number of scalar multiplications = nnz , which is potentially much less than n^2 .

See the TACO research project if you are interested (<https://tensor-compiler.org/>)

Sparsity (4)

- Storing a sparse graph



- Many graph algorithms run efficiently on this representation, e.g., breadth-first search, PageRank.

Outline

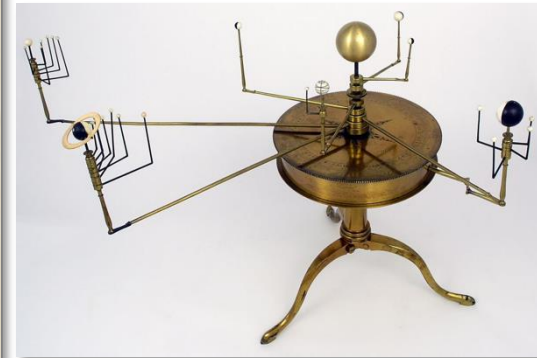
- Data structures
- **Logic**
- Loops
- Functions

Constant Folding and Propagation

- The idea of is to **evaluate** constant expressions, and **substitute** the result into further expressions, all during **compilation**.

```
#include <math.h>

void orrery() {
    const double radius = 6371000.0;
    const double diameter = 2 * radius;
    const double circumference = M_PI * diameter;
    const double cross_area = M_PI * radius * radius;
    const double surface_area =
        circumference * diameter;
    const double volume =
        4 * M_PI * radius * radius * radius / 3;
    // ...
}
```



mechanical orrery¹

With a sufficiently high optimization level, all the expressions are evaluated at compile-time.

¹[https://en.wikipedia.org/wiki/Orrery#/media/File:Thinktank_Birmingham_-_object_1956S00682.00001\(1\).jpg](https://en.wikipedia.org/wiki/Orrery#/media/File:Thinktank_Birmingham_-_object_1956S00682.00001(1).jpg)

Common-Subexpression Elimination

- The idea is to avoid computing the same expression multiple times by **evaluating** the expression **once** and **saving** the result for later use

```
a = b + c;  
b = a - d;  
c = b + c;  
d = a - d;
```

```
a = b + c;  
b = a - d;  
c = b + c;  
d = b;
```

Common-Subexpression Elimination

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a = b + c;  
b = a - d;  
c = b + c;  
d = a - d;
```

```
a = b + c;  
b = a - d;  
c = b + c;  
d = b;
```

The third line cannot be replaced by **c = a**,
because the value of **b** changes in the second line

Algebraic Identities

- The idea is to **replace** expensive algebraic expressions with algebraic equivalents that require less work

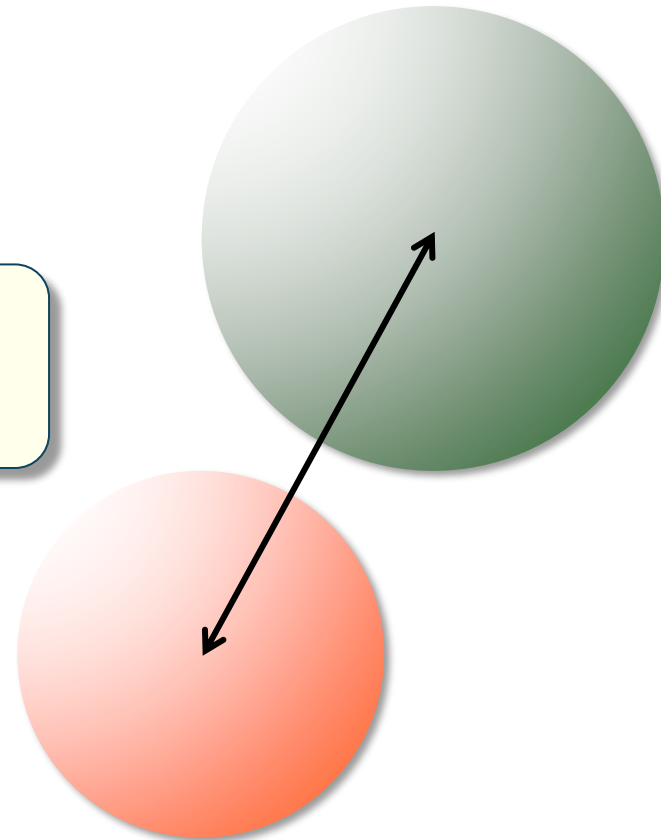
```
#include <stdbool.h>
#include <math.h>

typedef struct {
    double x, y, z; // spatial coordinates
    double r;       // radius of ball
} ball_t;

double square(double x) {
    return x*x;
}

bool collides(ball_t *b1, ball_t *b2) {
    double d = sqrt(square(b1->x - b2->x)
                    + square(b1->y - b2->y)
                    + square(b1->z - b2->z));
    return d <= b1->r + b2->r;
}
```

Expensive
routine!



Algebraic Identities

- The idea is to **replace** expensive algebraic expressions with algebraic equivalents that require less work

```
#include <stdbool.h>
#include <math.h>

typedef struct {
    double x, y, z; // spatial coordinates
    double r;       // radius
} ball_t;
```

```
double square(double x) {
    return x*x;
}
```

```
bool collides(ball_t *b1, ball_t *b2) {
    double d = sqrt(square(b1->x - b2->x)
                    + square(b1->y - b2->y)
                    + square(b1->z - b2->z));
    return d <= b1->r + b2->r;
}
```

$$\sqrt{u} \leq v \text{ exactly when } u \leq v^2$$

```
bool collides(ball_t *b1, ball_t *b2) {
    double dsquared = square(b1->x - b2->x)
                    + square(b1->y - b2->y)
                    + square(b1->z - b2->z);
    return dsquared <= square(b1->r + b2->r);
}
```

Caution: Be careful
with floating point!

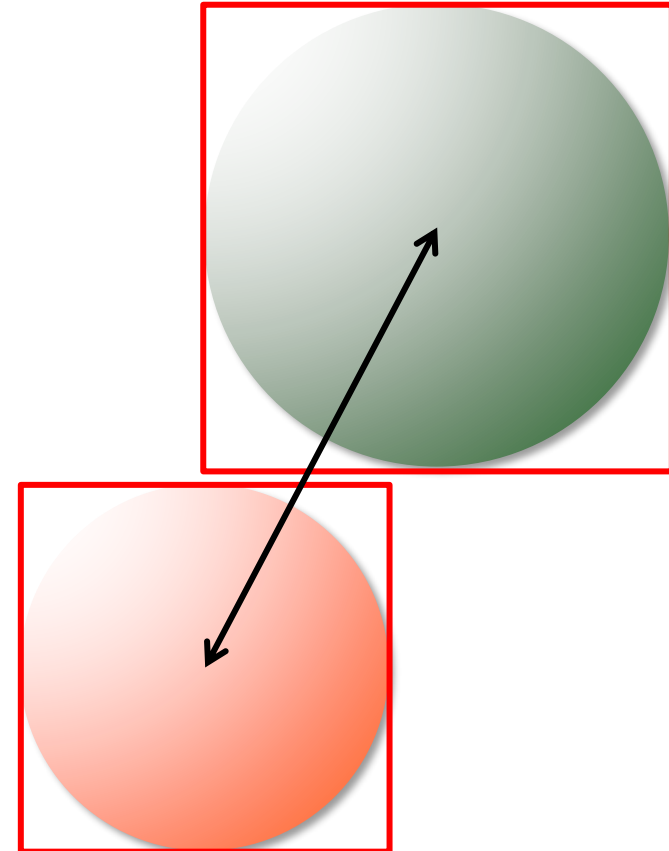
Creating a Fast Path

```
#include <stdbool.h>
#include <math.h>

typedef struct {
    double x, y, z; // spatial coordinates
    double r;       // radius of ball
} ball_t;

double square(double x) {
    return x*x;
}

bool collides(ball_t *b1, ball_t *b2) {
    double dsquared = square(b1->x - b2->x)
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                    + square(b1->z - b2->z);
    return dsquared <= square(b1->r + b2->r);
}
```



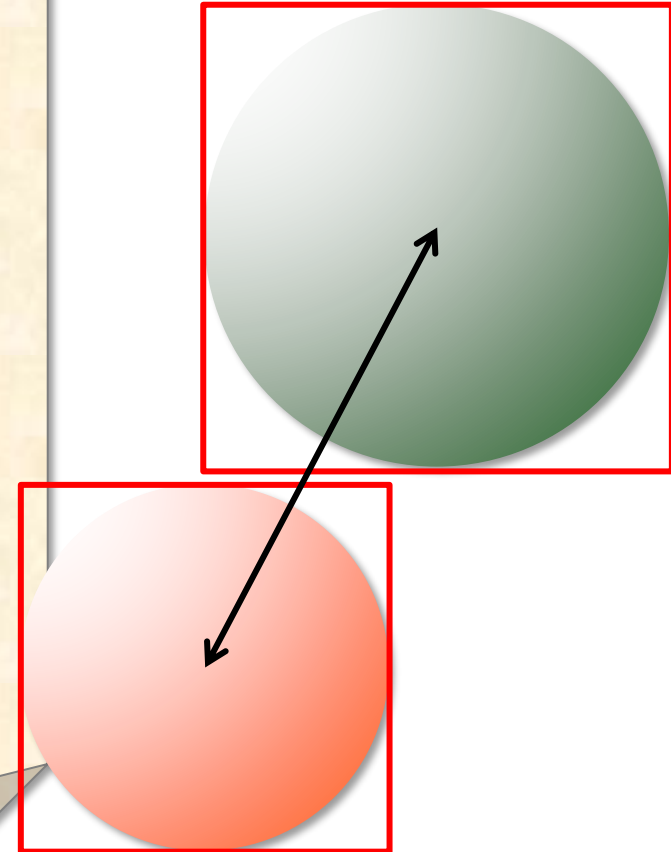
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    double dsquared = square(b1->x - b2->x)
                    + square(b1->y - b2->y)
                    + square(b1->z - b2->z);
    return dsquared <= square(b1->r + b2->r);
}
```



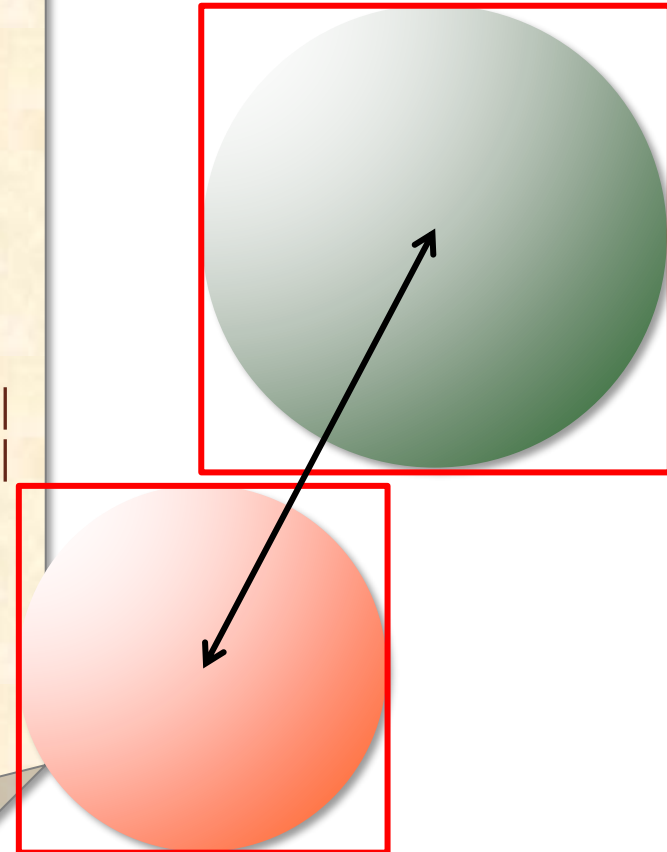
Creating a Fast Path

```
#include <stdbool.h>
#include <math.h>

typedef struct {
    double x, y, z; // spatial coordinates
    double r;       // radius of ball
} ball_t;

double square(double x) {
    return x*x;
}

bool collides(ball_t *b1, ball_t *b2) {
    if ((abs(b1->x - b2->x) > (b1->r + b2->r)) ||
        (abs(b1->y - b2->y) > (b1->r + b2->r)) ||
        (abs(b1->z - b2->z) > (b1->r + b2->r)))
        return false;
    double dsquared = square(b1->x - b2->x)
                      + square(b1->y - b2->y)
                      + square(b1->z - b2->z);
    return dsquared <= square(b1->r + b2->r);
}
```



Short-Circuiting

- The idea is to **stop** evaluating **as soon as** you know the answer, when performing a series of tests

```
#include <stdbool.h>
// All elements of A are nonnegative
bool sum_exceeds(int *A, int n, int limit) {
    int sum = 0;
    for (int i = 0; i < n; i++) {
        sum += A[i];
    }
    return sum > limit;
}
```

Short-Circuiting

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        sum += A[i];
    }
    return sum > limit;
}
```

Before

```
#include <stdbool.h>
// All elements of A are nonnegative
bool sum_exceeds(int *A, int n, int limit) {
    int sum = 0;
    for (int i = 0; i < n; i++) {
        sum += A[i];
        if (sum > limit) {
            return true;
        }
    }
    return false;
}
```

After

Ordering Tests

- Consider code that executes a sequence of logical tests.
- The idea is to perform those that are **more often “successful”** **before** tests that are **rarely successful**.

```
#include <stdbool.h>
bool is_whitespace(char c) {
    return (c == '\r' || c == '\t' || c == ' ' || c == '\n');
```

Before

```
#include <stdbool.h>
bool is_whitespace(char c) {
    return (c == ' ' || c == '\n' || c == '\t' || c == '\r');
```

After

Note that **&&** and **||** are short-circuiting logical operators, whereas **&** and **|** are not.

Combining Tests

- The idea is to **replace** a sequence of tests with one test or switch

Full adder

a	b	c	carry	sum
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

```
void full_add(int a,
              int b,
              int c,
              int *sum,
              int *carry) {
    if (a == 0) {
        if (b == 0) {
            if (c == 0) {
                *sum = 0;
                *carry = 0;
            } else {
                *sum = 1;
                *carry = 0;
            }
        } else {
            if (c == 0) {
                *sum = 1;
                *carry = 0;
            } else {
                *sum = 0;
                *carry = 1;
            }
        }
    }
}
```

```
} else {
    if (b == 0) {
        if (c == 0) {
            *sum = 1;
            *carry = 0;
        } else {
            *sum = 0;
            *carry = 1;
        }
    } else {
        if (c == 0) {
            *sum = 0;
            *carry = 1;
        } else {
            *sum = 1;
            *carry = 1;
        }
    }
}
```

Combining Tests (2)

- The idea is to **replace** a sequence of tests with one test or switch

Full adder

a	b	c	carry	sum
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

In this case, the outputs can be computed mathematically.

```
void full_add(int a,
              int b,
              int c,
              int *sum,
              int *carry) {
    int test = ((a == 1) << 2)
               | ((b == 1) << 1)
               | (c == 1);
    switch(test) {
        case 0:
            *sum = 0;
            *carry = 0;
            break;
        case 1:
            *sum = 1;
            *carry = 0;
            break;
        case 2:
            *sum = 1;
            *carry = 0;
            break;
```

```
        case 3:
            *sum = 0;
            *carry = 1;
            break;
        case 4:
            *sum = 1;
            *carry = 0;
            break;
        case 5:
            *sum = 0;
            *carry = 1;
            break;
        case 6:
            *sum = 0;
            *carry = 1;
            break;
        case 7:
            *sum = 1;
            *carry = 1;
            break;
```

```
    }
}
```

Outline

- Data structures
- Logic
- **Loops**
- Functions

Why Loops?

- **Loops** are often the focus of performance optimization. Why?

Loops account for a lot of work!

Consider this thought experiment:

- Suppose that a 2 GHz processor can execute 1 instruction per clock cycle.
- Suppose that a program contains 16 GB of instructions, but it is all **simple straight-line code**, i.e., no backwards branches.
- **Question:** How long does the code take to run?

Answer: at most 8 seconds!

What Happens When a Loop Runs?

Pseudocode for
loop execution

A simple loop

```
int sum = 0;
for (int i = 0; i < N; i++) {
    sum += A[i];
}
```



Loop
control

```
int sum = 0;
int i = 0;
if (i >= N)
    goto loop_exit;
sum += A[i];
i++;
if (i >= N)
    goto loop_exit;
sum += A[i];
i++;
if (i >= N)
    goto loop_exit;
sum += A[i];
i++;
if (i >= N)
    goto loop_exit;
// ...
```

Loop Unrolling

- The idea is to save work by **combining consecutive iterations** of a loop into a single iteration

- **Full** loop unrolling: All iterations are unrolled.

- **Partial** loop unrolling: Several, but not all, of the iterations are unrolled.

Loop unrolling **reduces** the total number of iterations of the loop and, consequently, the number of times that the instructions that control the loop must be executed.

Full Loop Unrolling

```
int sum = 0;
for (int i = 0; i < 10; i++) {
    sum += A[i];
}
```

Before

```
int sum = 0;
sum += A[0];
sum += A[1];
sum += A[2];
sum += A[3];
sum += A[4];
sum += A[5];
sum += A[6];
sum += A[7];
sum += A[8];
sum += A[9];
```

After

Partial Loop Unrolling

```
int sum = 0;
for (int i = 0; i < n; ++i) {
    sum += A[i];
}
```

Before

```
int sum = 0;
int j;
for (j = 0; j < n-3; j += 4) {
    sum += A[j];
    sum += A[j+1];
    sum += A[j+2];
    sum += A[j+3];
}
for (int i = j; i < n; ++i) {
    sum += A[i];
}
```

After

Benefits of loop unrolling

- Fewer instructions devoted to loop control.
- Enables more compiler optimizations.

Caution: Unrolling too much can cause poor use of the instruction cache, because the code is bigger.

Hoisting

- The idea is to avoid recomputing **loop-invariant** code each time through the body of a loop. **a.k.a. loop-invariant code motion**

```
#include <math.h>

void scale(double *X, double *Y, int N) {
    for (int i = 0; i < N; i++) {
        Y[i] = X[i] * exp(sqrt(M_PI/2));
    }
}
```

Before

```
#include <math.h>

void scale(double *X, double *Y, int N) {
    double factor = exp(sqrt(M_PI/2));
    for (int i = 0; i < N; i++) {
        Y[i] = X[i] * factor;
    }
}
```

After

Hoisting

- The idea is to avoid recomputing **loop-invariant** code each time through the body of a loop.

```
#include <math.h>

void scale(double *X, double *Y, int N) {
    for (int i = 0; i < N; i++) {
        Y[i] = X[i] * exp(sqrt(M_PI/N));
    }
}
```

Before

Loop Fusion

- The idea is to **combine multiple loops** over the same index range into a single loop body, thereby saving the overhead of loop control. a.k.a. **Jamming**

```
for (int i = 0; i < n; ++i) {  
    C[i] = (A[i] <= B[i]) ? A[i] : B[i];  
}  
  
for (int i = 0; i < n; ++i) {  
    D[i] = (A[i] <= B[i]) ? B[i] : A[i];  
}
```

Before

Ternary operator for
if-else.

```
for (int i = 0; i < n; ++i) {  
    C[i] = (A[i] <= B[i]) ? A[i] : B[i];  
    D[i] = (A[i] <= B[i]) ? B[i] : A[i];  
}
```

After

Eliminating Wasted Iterations

- The idea is to modify loop bounds to **avoid** executing loop iterations over essentially **empty loop bodies**.

<pre>for (int i = 0; i < n; ++i) { for (int j = 0; j < n; ++j) { if (i > j) { int temp = A[i][j]; A[i][j] = A[j][i]; A[j][i] = temp; } } }</pre>	Before
<pre>for (int i = 1; i < n; ++i) { for (int j = 0; j < i; ++j) { int temp = A[i][j]; A[i][j] = A[j][i]; A[j][i] = temp; } }</pre>	After

Outline

- Data structures
- Logic
- Loops
- **Functions**

Inlining

- The idea is to avoid the overhead of a function call by **replacing** a **call** to the function with the **body** of the function itself

```
double square(double x) {  
    return x*x;  
}
```

Before

```
double sum_of_squares(double *A, int n) {  
    double sum = 0.0;  
    for (int i = 0; i < n; ++i) {  
        sum += square(A[i]);  
    }  
    return sum;  
}
```

```
double sum_of_squares(double *A, int n) {  
    double sum = 0.0;  
    for (int i = 0; i < n; ++i) {  
        double temp = A[i];  
        sum += temp*temp;  
    }  
    return sum;  
}
```

After

Inlining (2)

- The idea is to avoid the overhead of a function call by **replacing** a **call** to the function with the **body** of the function itself

Ask the compiler
to inline for you.

`__attribute__((always_inline))`

```
inline double square(double x) {  
    return x*x;  
}  
  
double sum_of_squares(double *A, int n) {  
    double sum = 0.0;  
    for (int i = 0; i < n; ++i)  
        sum += square(A[i]);  
    return sum;  
}
```

Inlined functions can be just as efficient as macros, and they are safer to use and better structured.

Tail-Recursion Elimination

- It is to remove the overhead of **a recursive call** that **occurs as the last step of a function** F
 - The call is replaced with a branch to the top of the function F , and
 - Storage for local variables of F is reused by the erstwhile recursive call

```
void quicksort(int *A, int n) {  
    if (n > 1) {  
        int r = partition(A, n);  
        quicksort (A, r);  
        quicksort (A + r + 1, n - r - 1);  
    }  
}
```

Before

```
void quicksort(int *A, int n) {  
    while (n > 1) {  
        int r = partition(A, n);  
        quicksort (A, r);  
        A += r + 1;  
        n -= r + 1;  
    }  
}
```

After

Coarsening Recursion

- The idea is to **increase** the **size** of the **base case** and handle it with more efficient code that avoids function-call overhead.

```
void quicksort(int *A, int n) { Before  
    while (n > 1) {  
        int r = partition(A, n);  
        quicksort (A, r);  
        A += r + 1;  
        n -= r + 1;  
    }  
}
```

```
#define THRESHOLD 64 After  
void quicksort(int *A, int n) {  
    while (n > THRESHOLD) {  
        int r = partition(A, n);  
        quicksort (A, r);  
        A += r + 1;  
        n -= r + 1;  
    }  
    // insertion sort for small arrays  
    for (int j = 1; j < n; ++j) {  
        int key = A[j];  
        int i = j - 1;  
        while (i >= 0 && A[i] > key) {  
            A[i+1] = A[i];  
            --i;  
        }  
        A[i+1] = key;  
    }  
}
```

New Bentley Rules

Data structures

- Packing and encoding
- Augmentation
- Caching
- Precomputation
- Compile-time initialization
- Sparsity

Loops

- Loop unrolling
- Hoisting
- Loop fusion
- Eliminating wasted iterations

Logic

- Constant folding and propagation
- Common-subexpression elimination
- Algebraic identities
- Creating a fast path
- Short-circuiting
- Ordering tests
- Combining tests

Functions

- Inlining
- Tail-recursion elimination
- Coarsening recursion

Closing Advice

- Avoid **premature optimization**. First, get correct working code. Then optimize, preserving correctness by **regression testing**
- **Reducing the work** of a program does not necessarily decrease its running time, but it is a **good heuristic**
- Many optimizations involve **tradeoffs**. Use a **profiler** to see what code needs to be optimized. (See Homework 2.)
- The **compiler** automates many low-level optimizations, but not all

If you find interesting examples of work optimization, please let us know!